

ANSH THAPA

B.TECH

CU24250224

Experiment NO. I : Of Fundamental Image Processing Operations using Python and OpenCV

AIM

To develop a comprehensive understanding of fundamental image processing techniques by implementing image acquisition, storage, color space conversion, geometric transformations, and image enhancement operations using Python and OpenCV, thereby establishing the foundation for advanced computer vision applications.

Purpose

Image processing forms the core of modern computer vision systems used in domains such as autonomous driving, healthcare, surveillance, robotics, and industrial automation. Before performing advanced tasks like feature extraction, object detection, or image classification, it is to understand digital images are and transformed. In this experiment, students will implement fundamental image operations using Python and OpenCV. They will learn to read, write, convert, resize, and manipulate digital images while understanding how these operations influence image quality and prepare images for subsequent computer vision tasks.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

```
uploaded = files.upload()
img_path = list(uploaded.keys())[0]
print("Uploaded file:", img_path)
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving 2025_Nitro_Option_3840x2400.jpg to 2025_Nitro_Option_3840x2400.jpg
Uploaded file: 2025_Nitro_Option_3840x2400.jpg

```
img = cv2.imread(img_path)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

print("=== Image Properties ===")
print("Dimensions (H x W):", img.shape[0], "x", img.shape[1])
print("Number of Channels:", img.shape[2])
print("Data Type:", img.dtype)
print("Total Pixels/Size:", img.size, "elements")
```

```
=== Image Properties ===
Dimensions (H x W): 2400 x 3840
Number of Channels: 3
Data Type: uint8
Total Pixels/Size: 27648000 elements
```

```
plt.imshow(img_rgb)
plt.title("1. Original Image (RGB)")
plt.axis('off')
plt.show()
```

1. Original Image (RGB)



```
gray = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2GRAY)

plt.imshow(gray, cmap='gray')
plt.title("2. Grayscale Conversion")
plt.axis('off')
plt.show()
```

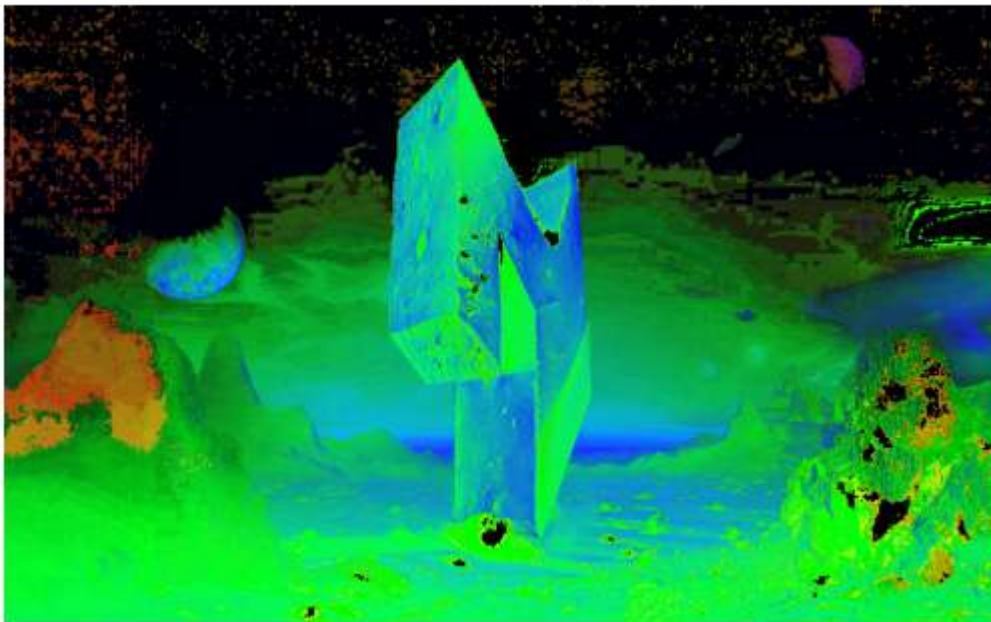
2. Grayscale Conversion



```
hsv = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2HSV)

plt.imshow(hsv)
plt.title("3. HSV Color Space")
plt.axis('off')
plt.show()
```

3. HSV Color Space



```
lab = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2LAB)

plt.imshow(lab)
plt.title("4. LAB Color Space")
plt.axis('off')
plt.show()
```

4. LAB Color Space



```
resized = cv2.resize(img_rgb, (256, 256))  
  
plt.imshow(resized)  
plt.title("5. Resized (256x256)")  
plt.axis('off')  
plt.show()
```

5. Resized (256x256)



```
rotated = cv2.rotate(img_rgb, cv2.ROTATE_90_CLOCKWISE)  
  
plt.imshow(rotated)  
plt.title("6. Rotated (-90°)")
```

```
plt.axis('off')  
plt.show()
```

6. Rotated (-90°)



```
flipped = cv2.flip(img_rgb, 1)  
  
plt.imshow(flipped)  
plt.title("7. Horizontal Flip")  
plt.axis('off')  
plt.show()
```

7. Horizontal Flip



```
negative = 255 - img_rgb

plt.imshow(negative)
plt.title("8. Negative (Inverted)")
plt.axis('off')
plt.show()
```

8. Negative (Inverted)



```
print(img_rgb.shape) # (Height, Width, Channels)
```

```
(2400, 3840, 3)
```

```
# Format: img[y1:y2, x1:x2]
cropped = img_rgb[100:2000, 1400:2400]

plt.imshow(cropped)
plt.title("9. Cropped ROI (Monolith)")
plt.axis('off')
plt.show()
```

9. Cropped ROI (Monolith)



```
titles = ['1. Original', '2. Grayscale', '3. HSV', '4. LAB',
          '5. Resized', '6. Rotated', '7. Flipped', '8. Negative', '9. Cropp
images = [img_rgb, gray, hsv, lab, resized, rotated, flipped, negative, crop

plt.figure(figsize=(15, 15))
for i in range(9):
    plt.subplot(3, 3, i+1)
    if len(images[i].shape) == 2:
        plt.imshow(images[i], cmap='gray')
    else:
        plt.imshow(images[i])
    plt.title(titles[i])
    plt.axis('off')

plt.tight_layout()
plt.show()
```



Answers to Experiment Questions

I. What is a digital image? between grayscale and images.

A digital image is a 2D matrix of discrete numerical elements called pixels, where each pixel value represents an intensity or color level at a specific spatial coordinate (x, y)

Grayscale vs. color:

Grayscale Represented by a single 2D channel (matrix), where pixel intensities range from 0 (pure black) to 255 (pure white) in standard 8-bit depth.

Color Images: Represented by a multi-channel 3D array (typically 3 channels: Red, Green, Blue), where a combination of the three channel intensities creates the full color palette.

2. Explain difference between RGB, BGR, HSV, and LAB color spaces

RGB (Red, Green, Blue): An additive color model standard in computer graphics, monitors, and web displays

BGR (Blue, Green, Red): Identical to RGB in structure, but with inverted channel ordering. Default format used internally by OpenCV.

HSV (Hue, Saturation, Value): Separates chromatic content from lightness. Hue defines the color angle (0° — 360°). Saturation represents color vibrancy/purity, Value represents lightness/brightness. Widely used for color-based thresholding and object tracking.

LAB (L, a, b): A perceptually uniform color space designed to mimic human vision. L represents Lightness, a defines the green-to-magenta spectrum, and b defines the blue-to-yellow spectrum. Separates luminance completely from color chrominance.

3. What is the purpose of converting an image to grayscale before further processing?

Grayscale conversion reduces a 3-channel color image to a single channel.

Its main purposes are:

- Reduces memory usage and computation.

- Makes algorithms like edge detection, thresholding and filtering easier.
- Removes unnecessary color variations and focuses on shapes, edges and texture.

4. Explain the concept of image complement (negative image) and mention its practical applications.

An image negative is obtained by subtracting each pixel value from the maximum intensity.

For an 8-bit image:

Negative = $255 - \text{Original Pixel}$

Thus, dark areas become bright and bright areas become dark.

Applications:

- Medical X-ray and CT image enhancement.
- Microscopy.
- Forensic analysis and fingerprint enhancement.

5. Differentiate between image resizing, cropping, and scaling.

- Resizing: Changes the overall width and height of an image.
- Cropping: Extracts a specific part of an image and removes the unwanted outer region.
- Scaling: Enlarges or shrinks an image using a scaling factor while generally maintaining its proportions.

6. What is a Region of Interest (ROI), and why is it important in computer vision?

ROI (Region of Interest) is a selected area of an image containing the specific object or region that needs to be analyzed.

Importance:

- Reduces processing time by focusing only on the required area.
- Removes unwanted background and noise.

- Improves detection and analysis accuracy.

7. Why is OpenCV preferred over conventional image processing libraries for computer vision applications?

OpenCV is preferred because:

- It has highly optimized C/C++ backend.
- Supports SIMD/AVX vectorization and multithreading.
- Provides hardware acceleration such as OpenCL and GPU support.
- Contains many ready-to-use computer vision algorithms.

8. Explain how image resolution and pixel intensity influence image quality.

Image resolution is the number of pixels in an image. Higher resolution provides more details, sharper edges and better representation of small objects.

Pixel intensity represents the brightness/value of a pixel. Higher bit depth provides a larger range of intensity values, producing better contrast and smoother gradients.

9. Mention five real-world applications where basic image preprocessing is an essential step.

1. Autonomous vehicles – lane detection and object recognition.
2. Medical imaging – X-ray, CT and MRI analysis.
3. OCR/Document scanning – text extraction and document correction.
4. Face recognition – face detection, alignment and resizing.
5. Biometric systems – fingerprint and facial verification.